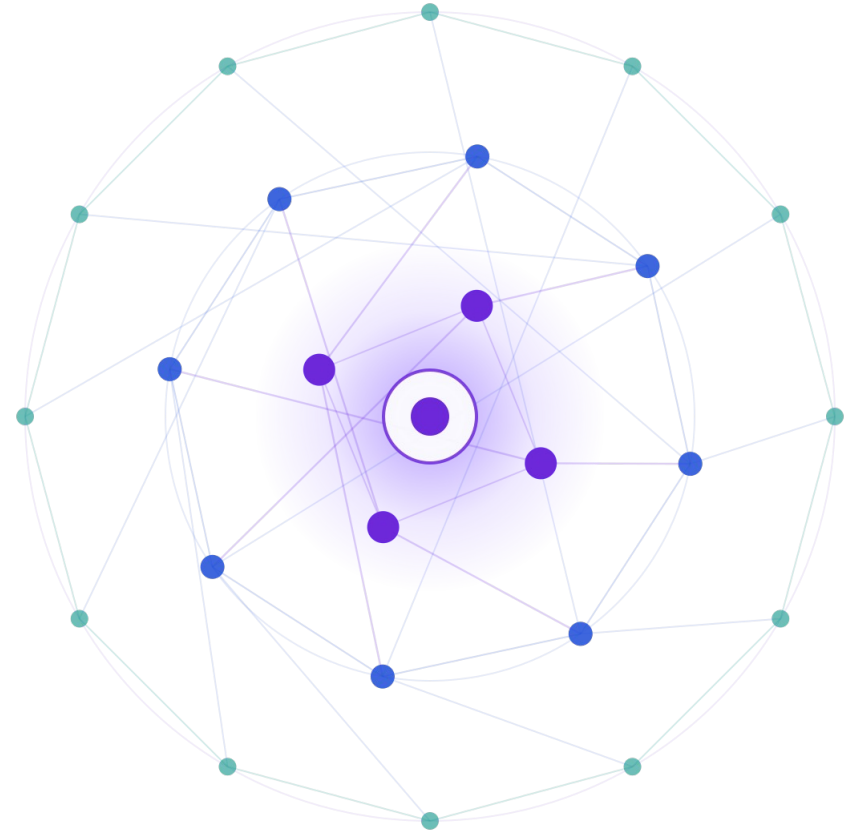


Generative AI Hackathon

3rd Edition 2026 – 27

- Theme ID - 05, Interruptible Real-Time Agents
- Project Title - An assistant you can interrupt mid-task
- Team Name - Santa Claude
- College Name - MSRIT
- Member Name & Email 1- Kartikey Pandey, Primary Data Analyst · [add email]
- Submission Github link - github.com/pandeykartik658-del/samsung-hackathon



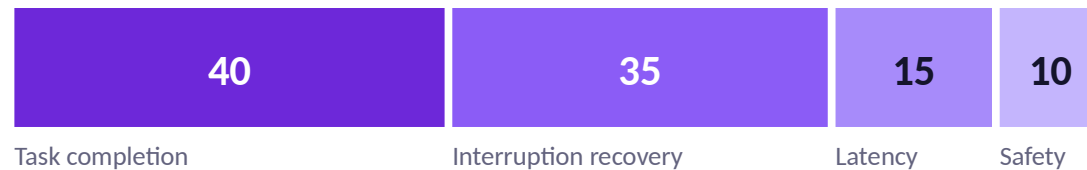
Theme

THE PROBLEM, IN OUR WORDS

Assistants take turns: listen, think, speak. People don't. They cut in, correct themselves mid-sentence and change the goal while a search or a booking is still running.

The agent has to keep talking, cancel the work nobody wants any more, re-plan on the new facts, and never repeat a side effect such as a booking.

HOW IT IS SCORED (THEME 05 GUIDE §5)



× quality multiplier 0.8–1.2. Audio and visual scenarios weigh 1.5×, and they are about half of the ~60 hidden scenarios.

FIVE THINGS A TURN-BASED BOT NEVER MEETS

- 1 **A correction lands** while the search it invalidates is still running.
- 2 **A result comes back** for a call the agent already cancelled.
- 3 **A booking must never run twice**, across retries, repeats and repairs.
- 4 **A tool it has never seen** arrives in the scenario manifest.
- 5 **Audio and camera frames** are slow next to the conversation.

Existing Solutions & Gaps

Half-duplex assistants

HOW IT WORKS

Listen, then think, then speak.

THE GAP

Go silent while they reason. An interruption only lands after the turn ends, so stale work keeps running.

Restart on interrupt

HOW IT WORKS

Throw the plan away and start over.

THE GAP

Repeat finished work, lose session context, and can send the same booking twice.

An LLM call on every step

HOW IT WORKS

Every decision waits on a model.

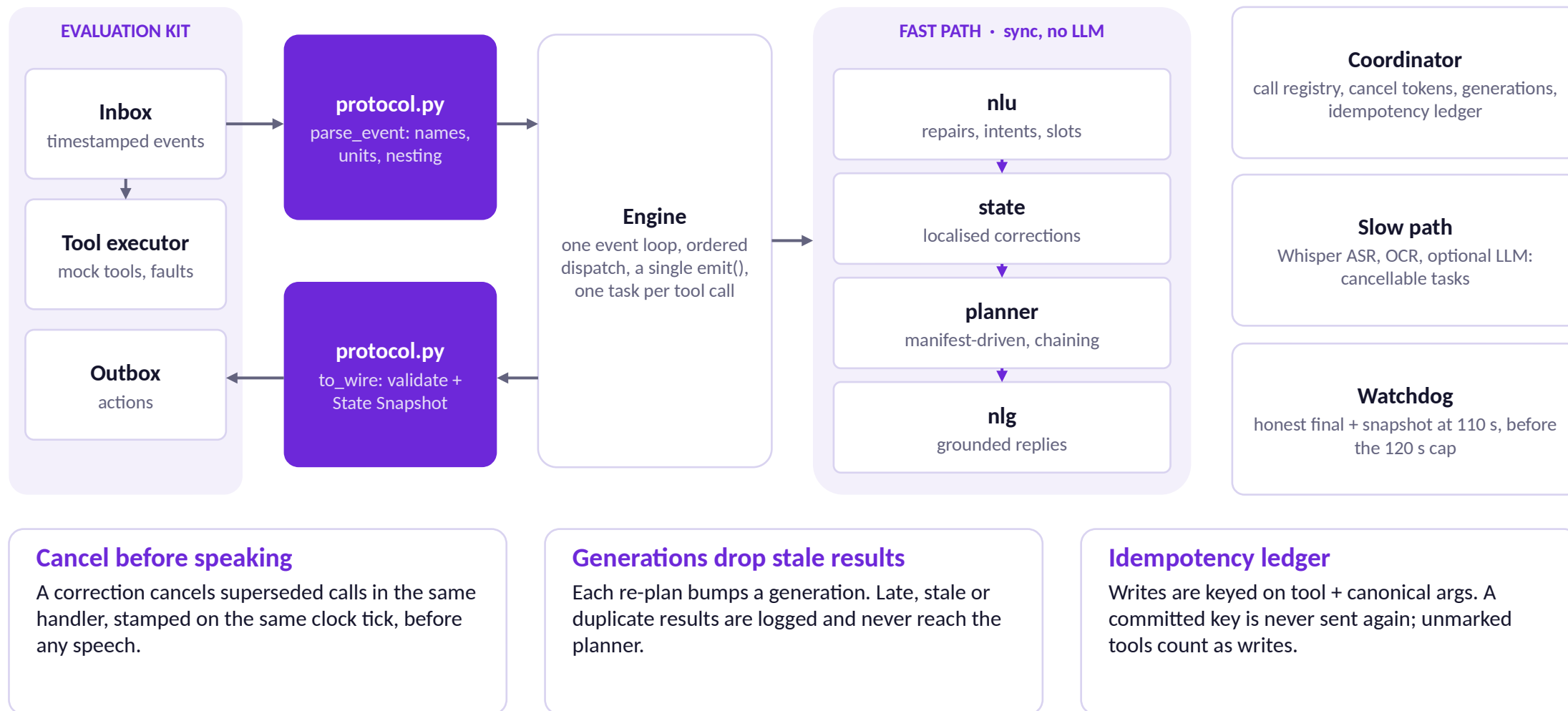
THE GAP

Seconds on the critical path, and no hard guarantee against duplicate writes.

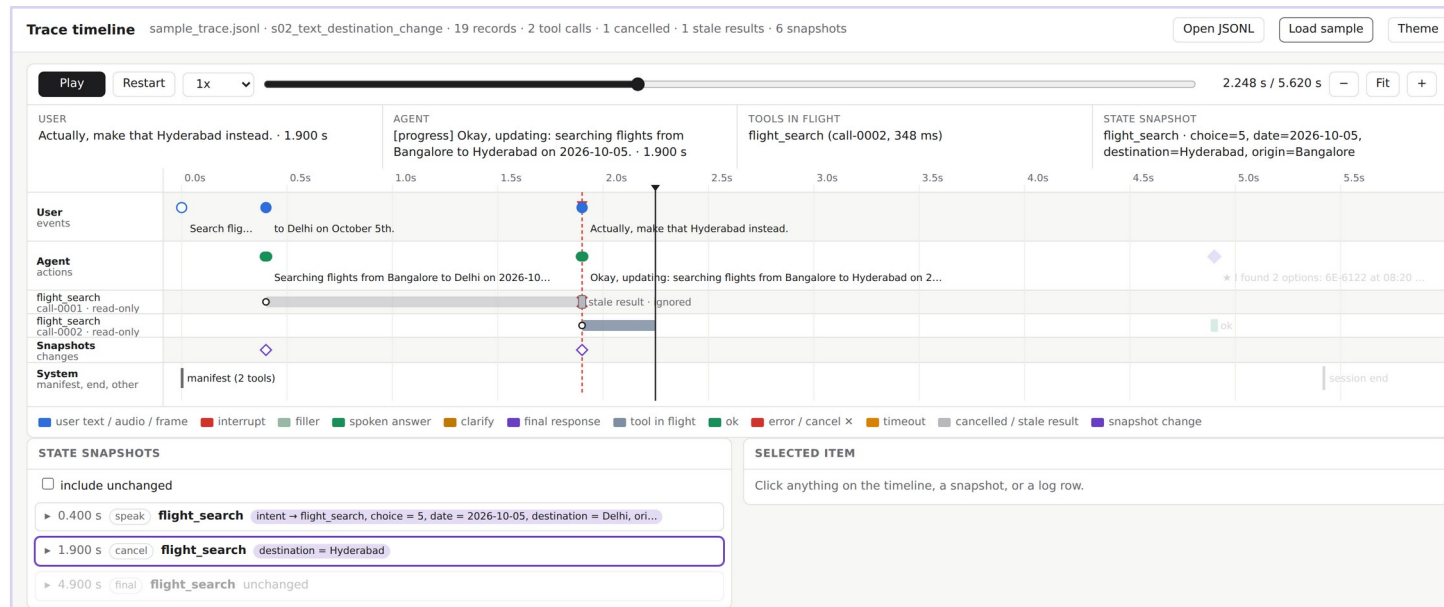
What we do instead

A rule-based fast path answers on the same event tick. A coordination layer cancels only the calls a correction invalidates. An idempotency ledger makes a second identical booking impossible.

Our Solutions & Architecture Diagram



Demo & Product Walkthrough



Trace viewer replaying scenario s02 on the virtual clock, paused just after the interruption.

WHAT THE DEMO VIDEO SHOWS

Destination change

cancel + re-plan

Duplicate-booking trap

one booking, not three

Ambiguous camera frame

asks, then answers

Video (max 5 min): [add YouTube or Drive link]

SCENARIO S02, SECOND BY SECOND

0.4 s

“...to Delhi on October 5th.”

Speaks a progress line, starts flight_search to Delhi.

1.9 s

“Actually, make that Hyderabad.”

Cancels the Delhi call on the same tick, then searches Hyderabad.

4.9 s

Hyderabad result arrives

Final answer with 2 flights and a snapshot: destination = Hyderabad.

Tools and tech stack used

RUNTIME

**Python 3.10–3.12,
asyncio**

Stdlib-only core. One Agent
per session, no module state.

SPEECH

**faster-whisper base.en,
int8**

Best CPU accuracy per second;
CTranslate2, no PyTorch.

VISION

RapidOCR (ONNX)

Reads display codes and
model labels. Weights ship in
the wheel.

LANGUAGE MODEL

Optional plug-in

2 s budget with a rule fallback.
Never on the fast path.

TESTING

pytest, 458 tests

Plus protocol fuzzing: 60,000
malformed events, zero
crashes.

EVALUATION

Own harness + scorer

Virtual-clock sim and a rubric
replica with 50 documented
assumptions.

DELIVERY

Docker, 1.01 GB

Whisper weights baked in,
runs offline. `python -m
theme5 serve` for JSONL.

CI AND DEMO

**GitHub Actions + trace
viewer**

Build, real-weights Whisper
test, all tests, demo. Single-file
HTML viewer.

Impact & Use case

In-car, hands-free

The destination changes while the route search runs.

The stale call is cancelled on the same tick; its late result is dropped.

Customer support

The customer changes details mid-booking.

The ledger blocks a second write; changes go through a modify tool or are refused.

Field troubleshooting

"How do I fix the error on this?" with the camera on.

Frames are grounded into device and error code; an unclear frame gets one question.

Accessibility

"Delhi, uh, no, Hyderabad." Hesitations and self-repairs.

Repair parsing changes only the corrected slot and keeps the rest.

Why it matters for Samsung devices: phones, cars and appliances need barge-in recovery without a network round trip. Our fast path answers in 0.69 ms (median) and runs offline.

Innovation highlights, results and limitations

100.0

60-scenario adversarial bench

87.4 on raw media only; 100.0 on a held-out seed

9 / 9

public-style scenarios pass

91.3 on raw media only (7 of 9 pass)

0.69 ms

median time to first spoken action

p95 1.27 ms

0

crashes in 60,000 fuzzed events

malformed, missing, out of order

All scores come from our own simulator and rubric replica. The official evaluation kit is not released yet.

INNOVATION HIGHLIGHTS

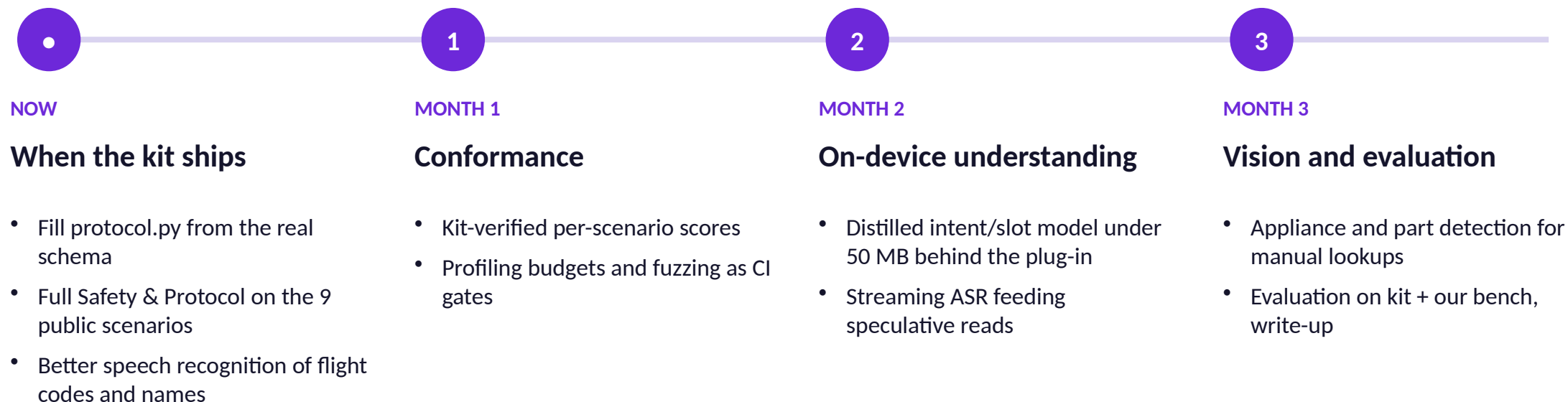
- Diff, don't flush: calls a correction leaves valid are kept, only the rest are cancelled.
- Speculative read-only calls on partial turns, adopted or cancelled at end of turn.
- Every kit guess sits in one adapter file, tagged ASSUMPTION with its unknown number.

LIMITATIONS

- 26 wire-format details are guesses until the kit ships.
- Self-graded, on scenarios we generated: a regression guard, not a forecast.
- Audio on raw media is the weak spot: most misses are garbled flight codes and passenger names. The agent asks instead of guessing, so it makes no wrong bookings. Frames on raw media score 100.
- Test audio and frames are synthetic. English only.

What's next

Worklet: a kit-conformant, learning-assisted interruptible agent for on-device assistants.



The rule stays: every new model is a plug-in with a timeout and a rule fallback, and the first action stays synchronous.

Brownie points slide(differentiation)

1

Replay any session

A single-file trace viewer plays a run on the virtual clock: calls, cancels, stale results, snapshots.

2

Never times out silently

A watchdog at 110 s cancels open work and gives one honest answer with a valid snapshot.

3

Adversarial bench

60 scenarios in 13 families (30 text, 18 audio, 12 visual), scored in about a second.

4

Ask, don't guess

Low ASR confidence, dark or ambiguous frames and missing parameters get one targeted question.

5

Unseen tools

The planner reads any manifest and binds arguments by name, type and enum; unmarked tools are treated as writes.

6

Fault tolerant

Retries only on retryable errors, and the ledger still blocks duplicate writes. Fuzzed with 60,000 malformed events.

Checklist- Updated on Public GitHub

- Working prototype code — public or shared GitHub repo (Y/N): Y, github.com/pandeykartik658-del/samsung-hackathon
- README with reproducible setup instructions (Y/N): Y, plus Dockerfile and CI
- Demo video, max 5 minutes (YouTube or Drive link): [add link]
- Presentation file (PPT or PDF) (Y/N): Y
- Release tag PRISM_GENAI_HACKATHON_Y2026 on the final commit: [to do]

Thank you

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India